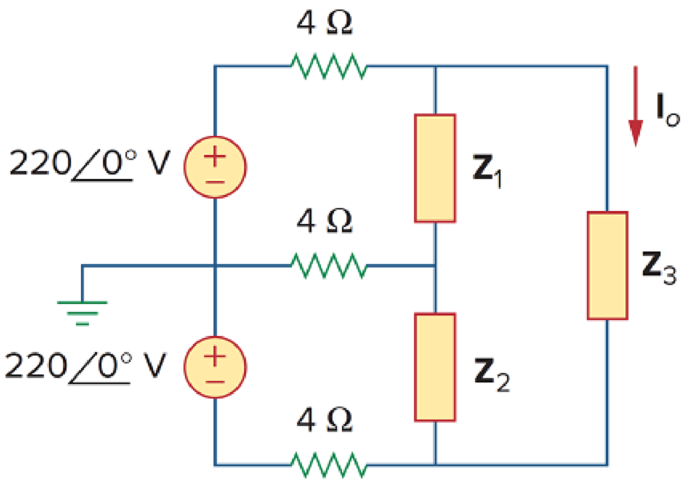
Chapter 12, Problem 60P

Problem

Use *PSpice or MultiSim* to determine **I***o* in the single-phase, three-wire circuit of Fig. 12.66. Let **Z**1 = 15 – *j*10 Ω, **Z**2 = 30 + *j*20 Ω, and **Z**3 = 12 + *j*5 Ω.

Figure 12.66



Step-by-step solution

Step 1 of 2

The schematics is shown below. IPRINT is inserted to give I0. We select

Total points = 1, Start frequency = 0.1892, End frequency = 0.1892 in

the sweep box, upon simulation, output file includes,

$Frequency$	$IN(V\_PRINT\,4)$	$IP(V\_PRINT\,4)$
$1.592\,E-01$	$1.421\,E+00$	$-1.355\,E+02$

Step 2 of 2

From which,

$$I_0 \quad = \quad 1.421\angle -135.5° \ A \quad .$$